Words and scents: How language promotes and skews olfactory processing

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Background: Language plays a paradoxical role in human olfaction. It can enhance odor categorization while simultaneously impairing recognition, suggesting that verbal influence varies across olfactory processes. Research shows that linguistic richness in smell vocabulary affects odor naming, evaluation, and memory (Ayabe-Kanamura et al., 1998; Majid & Burenhult, 2014; Wnuk et al., 2020). While verbal labels can scaffold odor categories, they may also distort recall and hedonic judgments (Herz & von Clef, 2001). This duality, where words support cognitive organization yet trigger olfactory illusions (Herz, 2005), aligns with the broader framework of the label-feedback hypothesis (Lupyan 2012). This study explores in three experiments how and in which contexts language facilitates or disrupts olfactory processing.

Exp. 1 Verbal facilitation of odor categories: Across four experimental groups, native Czech speakers underwent a four-day training to categorize hard-to-identify odors paired either with pseudowords or nonverbal sounds under high (81%) or low (25%) consistency conditions. When odors were paired with consistent pseudowords (*CONCORD*), participants showed the greatest improvement (M = 70.01%), outperforming those with inconsistent verbal pairings (*DISCORD:* M = 57.39%). In contrast, when odors were paired with structured nonverbal sounds, categorization gains were minimal in the *CONCORD* group (M = 55.49%) and unexpectedly higher in the DISCORD group (M = 64.58%). A GAMM test returned a significant difference in learning trajectories between the groups. These results confirm that verbal labels uniquely enhance odor category learning, beyond what nonverbal cues can achieve.

Exp. 2 Limits of verbally assisted odor categorization: Eighty Chinese-English bilinguals were trained over four days to categorize odors using pseudowords in Chinese, English, Georgian, or without labels. All groups improved significantly (ps < .001), but none of the verbal label conditions outperformed the Control group (ps > .219). Despite higher numerical gains in the Georgian group (29.38%), followed by English (25%) and Chinese (23.5%), Bayesian analysis corroborated these findings, showing moderate evidence for the null hypothesis across comparisons (e.g., BF_{10} = 0.15 for Chinese vs. Control). This indicates that verbal labels, regardless of language, did not reliably boost olfactory category learning, likely because added verbal processing load limited the formation of stable smell-word associations.

Exp. 3 Verbal skewing of odor memories: In this experiment, 140 Chinese participants were exposed to a target odor, completed a language distractor task, and were assigned to one of seven conditions differing in verbal description and response format. Those in verbal conditions described the odor before a multiple-choice recognition test with eight odors; while controls did not. Verbal description impaired recognition (Verbal Self-paced < Control Self-paced, p = .040), confirming that verbal overshadowing extends to odor recognition. This effect disappeared under time pressure (p = 1), suggesting speeded tasks reduce interference. Even when participants were asked to suppress or compare verbal and olfactory memory (Verbal Ignore, Verbal Two-choice), recognition remained lower (p = .021, p = .010).

Discussion: This study synthesizes findings from three experiments on olfactory categorization and verbal overshadowing, demonstrating that language plays a dual role in odor cognition, facilitating category learning while potentially distorting recognition. Verbal labels enhance category formation, in line with the label-feedback hypothesis (Lupyan, 2012), which suggests that labels guide attention toward category-relevant perceptual features. However, Experiment 2 showed that this facilitation is not guaranteed. When label-percept mappings are inconsistent or cognitively taxing, language may fail to boost learning. In recognition tasks, especially with hard-to-name odors, language can impair accuracy. These results deepen our understanding of how language can both scaffold and skew sensory perception, aligning closely with the 2026 HSP conference theme *Language in Minds and Machines* by showing how linguistic input dynamically modulates olfactory representations in the human mind.

Fig. 1 Exp. 1 design using (a) Sniffin' sticks as olfactory stimuli, (b) Bisyllabic pseudowords as verbal stimuli, and (c) odor categorization test without pseudowords

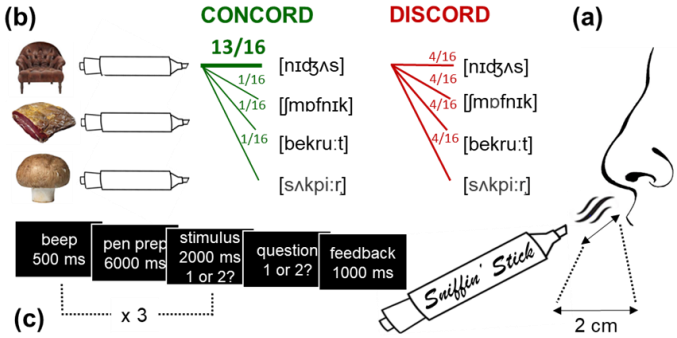


Fig. 2 Changes in categorization accuracy in the Concordant vs Discordant conditions per triplet (Exp. 1)

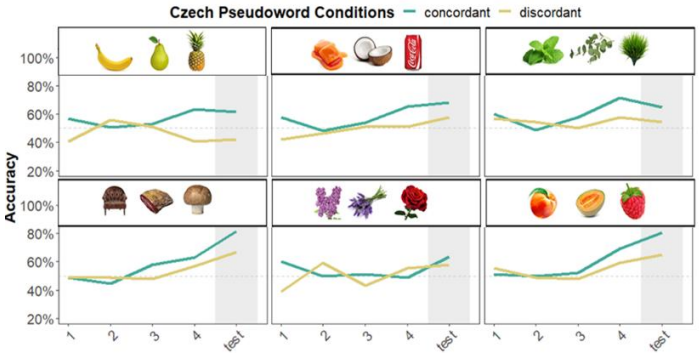


Fig. 3 Changes in categorization accuracy over four training sessions and the Test session per triplet (Exp. 2)

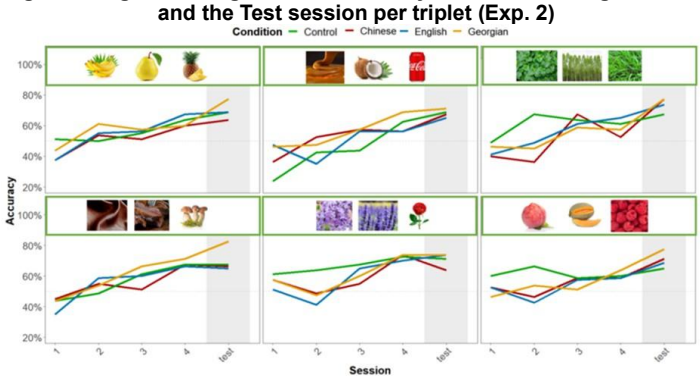


Fig. 4 Development of odor categorization accuracy per participant across training sessions and the test (Exp. 2)

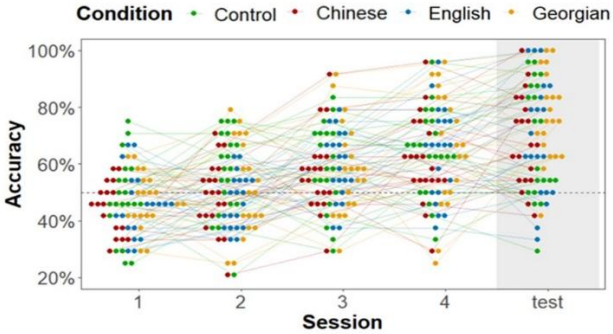


Fig. 5 Olfactory stimuli in the verbal overshadowing task (Exp.3)

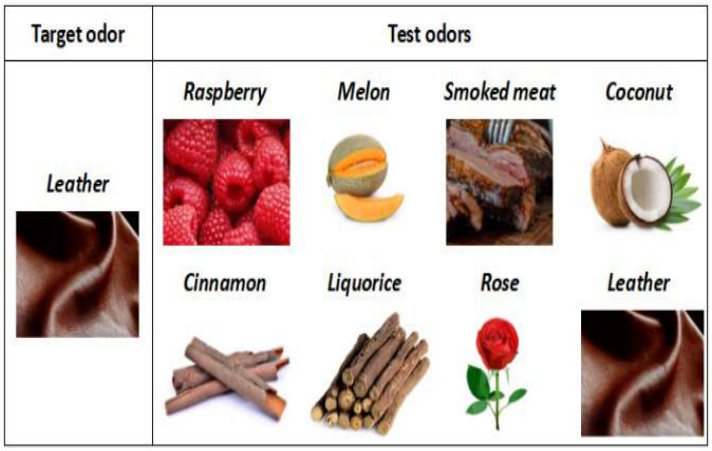
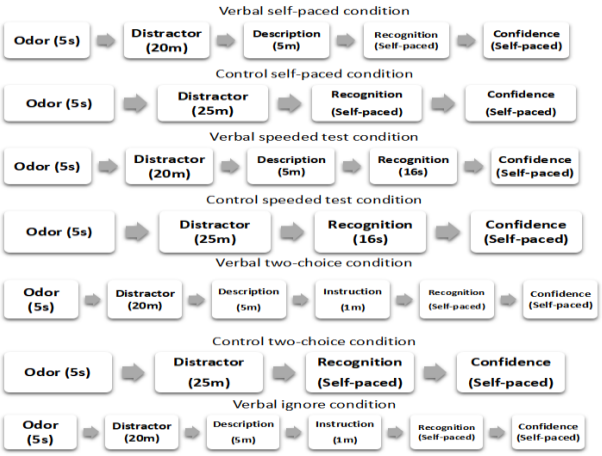


Fig. 6 Experimental procedures per condition (Exp. 3)



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